Numbers All Around Us: Rekenrek Subitizing & Ten-Structures

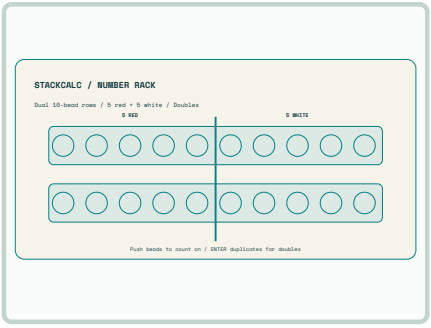
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Push 5 red beads and 3 white beads together in a single rapid slide.

Sketch Model on Paper:
Sketch the beads on a 2-wire diagram: 5 red + 3 white = 8.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	5	X: 5	5 red beads into X
2.	ENTER	Y: 5 | X: 5	Stack lift
3.	4	Y: 5 | X: 4	4 white beads into X
4.	+	X: 9	Add: 5 + 4 = 9

EXPECTED LCD X: 9 beads

1 **5 red beads and 3 white beads. Total?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **5 red beads and 4 white beads. Total?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **All 10 beads on wire. Slide 2 back right.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **All 10 beads on wire. Slide 5 back right.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 1

Numbers All Around Us: Rekenrek Subitizing & Ten-Structures • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Number Rack (20-Bead Dual Rod)

CAD Blueprint: number_rack

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White Frame with Signal Red & White Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**5 red beads and 3 white beads. Total?** Standard: CCSS.MATH.1.OA.C.6	8	[5] [ENTER] [3] [+]
2	**5 red beads and 4 white beads. Total?** Standard: CCSS.MATH.1.OA.C.6	9	[5] [ENTER] [4] [+]
3	**All 10 beads on wire. Slide 2 back right.** Standard: CCSS.MATH.1.OA.C.6	8	[1][0] [ENTER] [2] [-]
4	**All 10 beads on wire. Slide 5 back right.** Standard: CCSS.MATH.1.OA.C.6	5	[1][0] [ENTER] [5] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Look for subitizing: students should recognize 5 red instantly without 1-by-1 counts.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.